

Extended Kalman/UFIR Filters for UWB-Based Indoor Robot Localization Under Time-Varying Colored Measurement Noise

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Abstract—In indoor robot localization by using ultra-wideband (UWB), the extended Kalman filter (EKF)-based algorithms suffer from the colored measurement noise (CMN) that degrades the localization accuracy and causes the divergence. To overcome this issue, we develop a hybrid colored EKF and colored extended unbiased finite impulse response (EFIR) filter (cEKF/EFIR filter) employing measurement differences. We also develop this algorithm using a filter bank on merged averaging horizons to be adaptive to time-varying CMN and call it the adaptive EKF/EFIR (aEKF/EFIR) filter. Experimental testing is provided in UWB-based indoor mobile robot localization environments. It is shown that the end-to-end colored EKF/EFIR and aEKF/EFIR filtering algorithms have better performances than the EKF, EFIR filter, and their modifications for CMN.

Index Terms—Colored measurement noise, extended Kalman filter (EKF), extended UFIR (EFIR) filter, ultra-wideband (UWB) localization.

I. INTRODUCTION

ROBOTS in the modern world are used to fulfill complex and dangerous work, which cannot always be performed by humans [1], [2], such as the routine inspection of power transmission line systems [3]. Under the COVID-19 pandemic, mobile robots replace humans to accomplish remote monitoring of patients [4]. Domestic robots are programmed to clean up the space intelligently and efficiently. However, without an exact knowledge of the robot coordinates of the location, the task cannot be completed satisfactorily [5] and accurate

estimators are required [6]. Consequently, much attention has been paid in the last decades to increase the localization accuracy in a real-time manner [7].

Various approaches have been developed for indoor and outdoor localization. For example, the navigation system fusing the global positioning system (GPS) and global navigation satellite system (GLONASS) has been considered in [8] to organize navigation for a car-like robot. In [9], the BeiDou Navigation Satellite System (BDS) has been investigated. The problem with narrowband interference suppression for GPS navigation has been solved by some authors using the neural network (NN) technology. In [10], an NN-based predictor was proposed for GPS anti-jamming applications. Also the long short-term memory (LSTM) has been used to improve the seamless ability of the GPS/inertial navigation system (INS) system [11]. In [12], a hybrid multiple optimal Kalman filter (KF) and NN structure has been described. This list can be extended with a comment that GPS ground solutions are not always reliable in indoor environments.

To operate in GPS-denied environments, robots require other navigation methods. A promising low-cost approach employs radio frequency identification (RFID) tags and serves efficiently in many applications [13], [14]. In [15], the RFID-based localization was combined with phase unwrapping and nonlinear optimization. Another opportunity is offered by ultra-wideband (UWB)-based devices. An example is the indoor multisensor information fusion scheme proposed in [16] for food delivery robots and further improved in [17]. Extensive investigations [5], [18] have shown that the RFID- and UWB-based localization technologies can be highly efficient in GPS-denied environments. Furthermore, it was shown that the UWB is suitable for high accuracy localization, while the RFID-based approach should be used when a low-cost navigation is required over a massive number of the tags.

In hybrid navigation schemes, a critical question arises of how to organize optimal data fusion to improve the localization accuracy [19]. Most widely used fusion algorithms employ the KF [20], [21]. For example, a hybrid tightly coupled inertial navigation system (INS) and GPS structure (INS/GPS) proposed in [22] uses a multirate KF. The KF-based predictive navigation was developed for mobile robot optical localization in [23]. An application of the unscented KF (UKF) to vehicle localization using the INS/GPS integrated system was discussed in [24]. In [25], the cubature KF (CKF) has been modified for the

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multirate strong tracking has been considered. Note that the KF-based fusion relies on white Gaussian noise with known statistics and accurate knowledge about the initial location. Since it is not often the case in industrial environments, the KF may be unsuitable for accurate fusion [26], [27], [28]. A more robust data fusion can be organized using the unbiased finite impulse response (UFIR) filter proposed in [6], [29], and [30]. This filter ignores noise and initial values and typically provides better localization accuracy in uncertain industrial environments that have been demonstrated in [31] for RFID-based robot self-localization and in [32] for INS/UWB human localization with missing data. There was also designed a distributed hybrid particle/UFIR filter to mitigate the non-line-of-sight effects in wireless sensor networks [33].

It has to be remarked now that most of the existing localization algorithms still rely on white Gaussian noise, which is often in odds with harsh industrial environments. An efficient solution can be found if we view the measurement noise as colored and modify the model accordingly. The KF modification for the colored Gauss–Markov measurement noise (CMN) was originally obtained in [34]. Then, the KF and UFIR filters were modified for CMN in [35], employing the backward Euler method that has resulted in efficient filtering algorithms. The first experimental results obtained in [36] for INS/UWB-based quadrotor localization using the UFIR filter modified for CMN have shown much higher accuracy and stability. Further improvement can be achieved if we make the coloredness factor matrix (CFM) time-varying and adaptive to the operating environments that motivates our present work.

In this article, a hybrid extended KF (EKF) and extended UFIR (EFIR) filter for UWB-based indoor localization under CMN employing measurement differencing will be considered. The filter designed in such a way is called the “colored” EKF/EFIR (cEKF/EFIR) filter. It is then modified for time-varying CFM adapted to nonconstant environments using the interactive multimodel (IMM) method [37], [38] and called the adaptive EKF/EFIR (aEKF/EFIR) filter. Experimental testing is provided to compare the prediction accuracy of the cEKF/EFIR filter, aEKF/EFIR filter, EKF, “colored” EKF (cEKF), EFIR filter, and “colored” EFIR (cEFIR) filter. The main contributions are as follows.

- 1) The cEKF/EFIR filter design for CMN using measurement differences.
- 2) The aEKF/EFIR filter design by adapting the CFM to time-varying CMN.
- 3) Experimental evidence of better performances of the cEKF/EFIR and aEKF/EFIR filtering algorithms.

This article is organized as follows. The localization scheme using UWB is described in Section II. Section III discusses the state-space model and designs the cEKF/EFIR and aEKF/EFIR filtering algorithms for CMN. Experimental testing of the filters designed by the UWB-based indoor mobile robot localization is provided in Section IV, and Section V obtains the conclusions.

II. UWB-BASED INDOOR ROBOT LOCALIZATION

The indoor localization scheme employing UWB for the robot organized to fuse measured ranges from the UWB

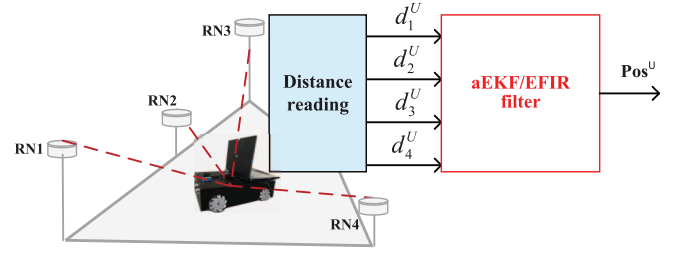


Fig. 1. UWB-based robot navigation scheme.

reference nodes (RNs) to a blind node (BN) has been proposed in [17] as shown in Fig. 1. The RNs are deployed in the indoor space with exactly known coordinates. The BN is mounted on a robot platform and is used to read distances d_i^U , $i \in [1, q]$, to the i th RN of q deployed RNs. In this article, we consider the localization system (Fig. 1) with $q = 4$. The cEKF/EFIR filter designed using the IMM method will be used to estimate the robot position Pos via the available distances d_i^U , $i \in [1, q]$, in the presence of CMN.

III. EKF/EFIR FILTERS DESIGN FOR UWB-BASED LOCALIZATION UNDER CMN

In this section, we will consider the state-space model and design the cEKF/EFIR and aEKF/EFIR filters to provide UWB-based indoor robot localization under CMN with constant and time-varying CFM adapted to the operation conditions.

A. State-Space Model

Following Fig. 1, the indoor robot localization system can be modeled in discrete-time state space if we introduce robot state vector $\mathbf{x}_k$ and a state equation

$$\begin{bmatrix} x_k \\ V_k^x \\ y_k \\ V_k^y \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{T} & \mathbf{0}_{2 \times 2} \\ \mathbf{0}_{2 \times 2} & \mathbf{T} \end{bmatrix}}_{\mathbf{F}} \begin{bmatrix} x_{k-1} \\ V_{k-1}^x \\ y_{k-1} \\ V_{k-1}^y \end{bmatrix} + \mathbf{w}_k \quad (1)$$

$\mathbf{x}_k^E$ $\mathbf{x}_{k-1}^E$

where k is the discrete time index; $\mathbf{T} = \begin{bmatrix} 1 & \tau \\ \tau & 1 \end{bmatrix}$; $\mathbf{0}_{2 \times 2} \in \mathbb{R}^{2 \times 2}$ is a matrix with all components zero; τ is the time step; (x_k, y_k) express the robot coordinates in the east direction x and north direction y , respectively; (V_k^x, V_k^y) are the robot velocities in the x and y directions, respectively; and $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$ is the white Gaussian system noise.

Referring to four RNs in Fig. 1 (the number of RNs can be arbitrary), the observation equation for measurements provided in the presence of the CMN can be written as

$$\underbrace{\begin{bmatrix} d_{1,k}^U \\ d_{2,k}^U \\ d_{3,k}^U \\ d_{4,k}^U \end{bmatrix}}_{\mathbf{y}_k} = \underbrace{\begin{bmatrix} \sqrt{(x_k - x_1)^2 + (y_k - y_1)^2} \\ \sqrt{(x_k - x_2)^2 + (y_k - y_2)^2} \\ \sqrt{(x_k - x_3)^2 + (y_k - y_3)^2} \\ \sqrt{(x_k - x_4)^2 + (y_k - y_4)^2} \end{bmatrix}}_{h(\mathbf{x}_k^E)} + \mathbf{v}_k \quad (2)$$

where $(x_i, y_i), i \in [1, 4]$, are coordinates of the i th RN in the (x, y) plane and $\mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$ is the white Gaussian measurement noise. With (1) and (2) being provided, the standard EKF and EFIR filters can be applied as shown in [17].

B. cEKF/EFIR Filtering Algorithm Under CMN

Since the white Gaussian assumption is not often the case for industrial applications and noise $\mathbf{v}_k$ is typically colored [36], we next consider $\mathbf{v}_n = [v_{d1}^U \ v_{d2}^U \ v_{d3}^U \ v_{d4}^U]^T$ as Markov–Gauss CMN and represent the state-space model with

$$\begin{cases} \mathbf{x}_k^{cE} = \mathbf{F}\mathbf{x}_{k-1}^{cE} + \mathbf{w}_k \\ \mathbf{y}_k = h(\mathbf{x}_k^{cE}) + \mathbf{v}_k \\ \mathbf{v}_k = \gamma\mathbf{v}_{k-1} + \varepsilon_k \end{cases} \quad (3)$$

where γ is the CFM, in which components are chosen to be positive and less than unity for the sake of stability. To modify the EKF and EFIR filters for CMN, we follow [34] and consider the measurements differences that yield a new observation

$$\begin{aligned} \mathbf{z}_k &= \mathbf{y}_k - \gamma\mathbf{y}_{k-1} = h(\mathbf{x}_k^{cE}) + \mathbf{v}_k - \gamma h(\mathbf{x}_{k-1}^{cE}) - \gamma\mathbf{v}_{k-1} \\ &= \underbrace{h(\mathbf{x}_k^{cE}) - \gamma h(\mathbf{x}_{k-1}^{cE})}_{\bar{h}(\mathbf{x}_k^{cE})} + \mathbf{v}_k - \gamma\mathbf{v}_{k-1} \\ &= \bar{h}(\mathbf{x}_k^{cE}) + \varepsilon_k. \end{aligned} \quad (4)$$

By expanding the nonlinear function $h(\mathbf{x}_k^{cE})$ with the first-order Taylor series as [31]

$$\begin{aligned} h(\mathbf{x}_k^{cE}) &\cong h(\hat{\mathbf{x}}_k^{cE}) + \frac{\partial h}{\partial \mathbf{x}} \Big|_{\hat{\mathbf{x}}_k^{cE}} (\mathbf{x}_k^{cE} - \hat{\mathbf{x}}_k^{cE}) \\ &\cong \mathbf{H}_k \mathbf{x}_k^{cE} \end{aligned} \quad (5)$$

where $\mathbf{H}_k = (\partial h / \partial \mathbf{x})|_{\hat{\mathbf{x}}_k^{cE}}$ is the Jacobian matrix, (4) can further be approximated with

$$\begin{aligned} \mathbf{z}_k &\cong \mathbf{H}_k \mathbf{x}_k^{cE} - \gamma \mathbf{H}_{k-1} \mathbf{x}_{k-1}^{cE} + \varepsilon_k \\ &= \mathbf{C}_k \mathbf{x}_k^{cE} + \bar{\mathbf{v}}_k \end{aligned} \quad (6)$$

where $\mathbf{C}_k = \mathbf{H}_k - \gamma \mathbf{H}_{k-1}$, $\bar{\mathbf{v}}_k = \Omega_k \mathbf{w}_k + \varepsilon_k$ is white Gaussian noise with the covariance

$$\bar{\mathbf{R}} = E\{\bar{\mathbf{v}}_k \bar{\mathbf{v}}_k^T\} = \Omega_k \mathbf{Q} \Omega_k^T + \mathbf{R} \quad (7)$$

and we notice that $\bar{\mathbf{v}}_k$ and $\mathbf{w}_k$ appear to be time-correlated.

To design the EKF, we define the prior error covariance as

$$\mathbf{P}_k^{cE} = \mathbf{F} \mathbf{P}_{k-1}^{cE} \mathbf{F}^T + \mathbf{Q} \quad (8)$$

and the innovation covariance $\mathbf{S}_k$ as

$$\begin{aligned} \mathbf{S}_k &= E\{(\mathbf{z}_k - \bar{h}(\mathbf{x}_k^{cE}))(\mathbf{z}_k - \bar{h}(\mathbf{x}_k^{cE}))^T\} \\ &\approx E\{(\mathbf{z}_k - \mathbf{C}_k \mathbf{x}_k^{cE})(\mathbf{z}_k - \mathbf{C}_k \mathbf{x}_k^{cE})^T\} \\ &= \mathbf{C}_k \mathbf{F} \mathbf{P}_{k-1}^{cE} \mathbf{F}^T \mathbf{C}_k^T + \mathbf{C}_k \mathbf{Q} \mathbf{C}_k^T + \Omega_k \phi_k \\ &\quad + \mathbf{R} + \mathbf{C}_k \phi_k + \phi_k^T \mathbf{C}_k^T \\ &= \mathbf{C}_k \mathbf{P}_k^- \mathbf{C}_k^T + \mathbf{R} + \mathbf{C}_k \phi_k + \phi_k^T \mathbf{C}_k^T \end{aligned} \quad (9)$$

where $\phi_k = \mathbf{Q} \Omega_k^T$. We next rewrite (1) as

$$\begin{aligned} \mathbf{x}_k^{cE} &= \mathbf{F} \mathbf{x}_{k-1}^{cE} + \mathbf{w}_k + \chi [\mathbf{z}_k - \bar{h}(\mathbf{x}_k^{cE}) - \varepsilon_k] \\ &\cong \mathbf{A}_k \mathbf{x}_{k-1}^{cE} + \mu_k + \delta_k \end{aligned} \quad (10)$$

Algorithm 1: cEKF/EFIR Algorithm

Data: $\mathbf{y}_k, \hat{\mathbf{x}}_0^{cE}, \mathbf{R}, \mathbf{Q}, M^{cE}, N^{cE}$
Result: $\hat{\mathbf{x}}_k^{cE}$

```

1 begin
2   for  $k = 1 : \infty$  do
3     if  $k < N - 1$  then
4        $\mathbf{z}_k = \mathbf{y}_k - \gamma \mathbf{y}_{k-1}$ ;
5        $\mathbf{C}_k = \mathbf{H}_k - \gamma \mathbf{H}_{k-1} \mathbf{F}^{-1}$ ;
6        $\hat{\mathbf{P}}_k^{cE} = \mathbf{A} \hat{\mathbf{P}}_{k-1}^{cE} \mathbf{A}^T + \Theta$ ;
7        $\mathbf{S}_k = \mathbf{C}_k \hat{\mathbf{P}}_k^{cE} \mathbf{C}_k^T + \mathbf{R} + \Omega_k \phi_k$ ;
8        $\mathbf{K}_k = \hat{\mathbf{P}}_k^{cE} \mathbf{C}_k^T \mathbf{S}_k^{-1}$ ;
9        $\hat{\mathbf{x}}_k^{cE} = \mathbf{A} \hat{\mathbf{x}}_{k-1}^{cE} + \chi \mathbf{z}_k$ ;
10       $\hat{\mathbf{x}}_k^{cE} = \hat{\mathbf{x}}_k^{cE} + \mathbf{K}_k [\mathbf{z}_k - h(\hat{\mathbf{x}}_k^{cE}) + \gamma h(\hat{\mathbf{x}}_{k-1}^{cE})]$ ;
11       $\hat{\mathbf{P}}_k^{cE} = (\mathbf{I} - \mathbf{K}_k \mathbf{C}_k) \hat{\mathbf{P}}_k^{cE}$ ;
12    else
13       $m^{cE} = k - N^{cE} + 1, \quad s = m^{cE} + M^{cE} - 1$ ;
14       $\tilde{\mathbf{x}}_s^{cE} = \hat{\mathbf{x}}_s^{cE}$ ;
15       $\mathbf{G}_l = \mathbf{I}$ ;
16      for  $l = s + 1 : k$  do
17         $\mathbf{z}_l = \mathbf{y}_l - \gamma \mathbf{y}_{l-1}$ ;
18         $\mathbf{C}_l = \mathbf{H}_l - \gamma \mathbf{H}_{l-1} \mathbf{F}^{-1}$ ;
19         $\mathbf{G}_l = [\mathbf{C}_l^T \mathbf{C}_l + (\mathbf{F} \mathbf{G}_{l-1} \mathbf{F}^T)^{-1}]^{-1}$ ;
20         $\mathbf{K}_l = \mathbf{G}_l \mathbf{C}_l^T$ ;
21         $\tilde{\mathbf{x}}_l^{cE} = \mathbf{F} \tilde{\mathbf{x}}_{l-1}^{cE}$ ;
22         $\tilde{\mathbf{x}}_l^{cE} = \tilde{\mathbf{x}}_l^{cE} + \mathbf{K}_l (\mathbf{z}_l - h(\tilde{\mathbf{x}}_l^{cE}) + \gamma h(\tilde{\mathbf{x}}_{l-1}^{cE}))$ ;
23      end for
24       $\hat{\mathbf{x}}_l^{cE} = \tilde{\mathbf{x}}_l^{cE}$ ;
25    end if
26  end for
27 end

```

† M^{cE} means $\hat{\mathbf{x}}_k^{cE}$'s size
 † N^{cE} means the horizon length
 † $\mathbf{H}_k = \frac{\partial h(\mathbf{x}_k^{cE})}{\partial \mathbf{x}_k^{cE}} \Big|_{\hat{\mathbf{x}}_k^{cE}}$

where $\mathbf{A}_k = (\mathbf{I} - \chi_k \mathbf{C}_k) \mathbf{F}$, $\mu_k = \chi_k \mathbf{z}_k$, and $\delta_k = (\mathbf{I} - \chi_k \mathbf{C}_k) \mathbf{w}_k - \chi_k \bar{\mathbf{v}}_k \sim \mathcal{N}(\mathbf{0}, \Theta)$ is white Gaussian with the covariance

$$\begin{aligned} \Theta &= E\{\delta_k \delta_k^T\} \\ &= E\{[(\mathbf{I} - \chi_k \mathbf{C}_k) \mathbf{w}_k - \chi_k \bar{\mathbf{v}}_k][\dots]^T\} \\ &= (\mathbf{I} - \chi_k \mathbf{C}_k) \mathbf{Q} (\mathbf{I} - \chi_k \mathbf{C}_k)^T + \chi_k \mathbf{R} \chi_k^T. \end{aligned} \quad (11)$$

It now needs to de-correlate δ_k and $\bar{\mathbf{v}}_k$ in order for the EKF algorithm to be suboptimal. That can be done if we follow [35], constrain the noise vectors by

$$\begin{aligned} E\{\delta_k \bar{\mathbf{v}}_k^T\} &= E\{[(\mathbf{I} - \chi_k \mathbf{C}_k) \mathbf{w}_k - \chi_k \bar{\mathbf{v}}_k] \\ &\quad \times (\mathbf{w}_k^T \Omega_k^T + \delta_k)\} = 0 \end{aligned} \quad (12)$$

and obtain

$$\begin{aligned} \chi_k &= \phi_k (\mathbf{H}_k \phi_k + \mathbf{R})^{-1} \\ \Theta &= (\mathbf{I} - \chi_k \mathbf{H}_k) \mathbf{Q} (\mathbf{I} - \chi_k \mathbf{C}_k)^T. \end{aligned} \quad (13)$$

Having the uncorrelated noise vectors, we modify the EKF and EFIR filters for CMN and list a pseudocode of the cEKF/EFIR filter as Algorithm 1.

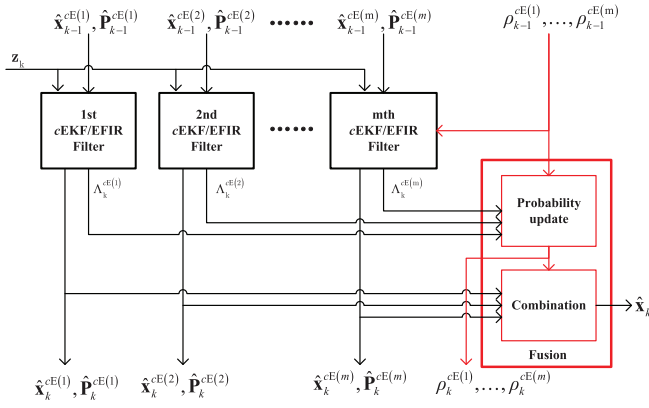


Fig. 2. Structure of the blind adaptive aEKF/EFIR filtering algorithm.

C. aEKF/EFIR Filtering Algorithm Under CMN

We now look into the robot dynamics in uncertain industrial environments and observe that the CMN may not be constant on finite horizons and thus the CFM γ should be time varying. Accordingly, we modify the EKF/EFIR filter to be adaptive as aEKF/EFIR. Such a filter will be blind if we provide the design as follows.

- 1) Select m candidate CFM γ_i , $i \in [1, m]$.
- 2) Tune the bank of m cEKF/EFIR subfilters to operate with γ_i , $i \in [1, m]$.
- 3) Fuse the subfilter outputs in the aEKF/EFIR filter using probabilistic weights.

A structure of the aEKF/EFIR filtering algorithm is shown in Fig. 2. To fuse optimally estimates $\hat{\mathbf{x}}_k^{cE(i)}$, $i \in [1, m]$, produced by the filter bank, we employ the Markov transition probability matrix

$$\Pi = \begin{bmatrix} \pi_{11} & \pi_{12} & \cdots & \pi_{1m} \\ \pi_{21} & \pi_{22} & \cdots & \pi_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ \pi_{m1} & \pi_{m2} & \cdots & \pi_{mm} \end{bmatrix} \quad (14)$$

where π_{ij} , $\{i, j\} \in [1, m]$, is the transition probability between the i th and j th outputs of the filter bank. Matrix ρ_k combines the time-varying weights $\rho_k^{cE(i)}$, $i \in [1, m]$, assigned for the subfilters as

$$\rho_k = \begin{bmatrix} \rho_k^{cE(1)} \\ \rho_k^{cE(2)} \\ \vdots \\ \rho_k^{cE(m)} \end{bmatrix}. \quad (15)$$

The probabilistic weighting matrix $\Lambda_k^{cE(i)} \triangleq \Lambda_k^{cE(i)}(\mathbf{z}_k | \hat{\mathbf{x}}_k^{cE(i)})$ is computed by (16), shown at the bottom of the page, where the i th weight is $\rho_k^{cE(i)} = \mathbf{y}_k - h(\hat{\mathbf{x}}_k^{cE(i)}) +$



Fig. 3. Experimental set organized to provide UWB-based indoor robot localization (barrier-free flat layer).

$\gamma h(\mathbf{x}_{k-1}^{cE(i)})$. Consequently, the weight $\rho_k^{cE(i)}$ is obtained by

$$\rho_k^{cE(i)} = \frac{1}{b_k} \Lambda_k^{cE(i)} \bar{\rho}_k^{cE(i)} \quad (17)$$

where $b_k = \sum_{i=1}^m \Lambda_k^{cE(i)} \bar{\rho}_k^{cE(i)}$ and $\bar{\rho}_k^{cE(i)} = \sum_{i=1}^m \pi_{1ik} \rho_k^{cE(i)}$.

Finally, the output of the aEKF/EFIR filter is provided by

$$\hat{\mathbf{x}}_k^{cE} = \sum_{i=1}^m \rho_k^{cE(i)} \hat{\mathbf{x}}_k^{cE(i)} \quad (18)$$

as a linear combination of the filter bank weighted outputs.

IV. EXPERIMENTAL TESTING

In this article, we experimentally test the cEKF/EFIR and aEKF/EFIR filters designed along with the EKF, EFIR filter, cEKF, and cEFIR filter using the indoor scene and implementing the scheme given in Fig. 1.

A. Test of Barrier-Free Environment

The experimental set used to test the filtering algorithms is pictured in Fig. 3, where we exploit the RoboMaster UWB localization system. Four UWB RNs are deployed at the corners of a (6×12) m² area and the UWB BN is mounted on a robot. Here, we select the positioning system using DW1000 as the UWB system.

To reach the best localization performance, all filters are tuned at a test stage in the best way to the model and noise. To make it near optimally, we consider model (3) with $M^{cE} = 4$ and set $\tau = 0.02$ s and $\sigma_w = 0.6$ m/s. Consequently, the noise

$$\Lambda_k^{cE(i)} = \frac{\exp \left[-\frac{1}{2} \rho_k^{cE(i)} \left(\mathbf{C}_k \mathbf{P}_k^{cE(i)} \mathbf{C}_k^T + \mathbf{R} + \mathbf{H}_k \phi_k + \phi_k^T \mathbf{C}_k^T \right)^{-1} \rho_k^{cE(i)} \right]}{\sqrt{2\pi \left(\mathbf{C}_k \mathbf{P}_k^{cE(i)} \mathbf{C}_k^T + \mathbf{R} + \mathbf{H}_k \phi_k + \phi_k^T \mathbf{C}_k^T \right)}} \quad (16)$$

TABLE I
PARAMETERS USED IN TWO EXPERIMENTAL TESTS

	$\mathbf{x}_0$	$\mathbf{P}_0$	$\mathbf{Q}$	$\mathbf{R}$
Test 1	$\mathbf{0}_{4 \times 1}$	$\mathbf{I}_{4 \times 4}$	$\mathbf{Q}$	$\mathbf{R}$
Test 2	$\mathbf{0}_{4 \times 1}$	$\mathbf{I}_{4 \times 4}$	$\mathbf{Q}/9$	$9\mathbf{R}$

covariance $\mathbf{Q}$ is composed as

$$\mathbf{Q} = (0.6 \times 0.2)^2 \times \begin{bmatrix} \frac{0.02^2}{4} & \frac{0.02}{2} & 0 & 0 \\ \frac{0.02}{2} & 1 & 0 & 0 \\ 0 & 0 & \frac{0.02^2}{4} & \frac{0.02}{2} \\ 0 & 0 & \frac{0.02}{2} & 1 \end{bmatrix}. \quad (19)$$

Following [39] and [40], the optimal scalar coloredness factor is measured as $\gamma_{\text{opt}} = 0.7$ and the suboptimal averaging horizon as $N^{cE} = 15$. From the test data, we also find the noise variances in the measured distances to be $\sigma_{v_{d_1^U}}^2 = 6.53 \text{ m}^2$, $\sigma_{v_{d_2^U}}^2 = 10.32 \text{ m}^2$, $\sigma_{v_{d_3^U}}^2 = 4.45 \text{ m}^2$, and $\sigma_{v_{d_4^U}}^2 = 10.51 \text{ m}^2$ and assign the measurement noise covariance $\mathbf{R}$ as

$$\mathbf{R} = \begin{bmatrix} 6.5322 & 0 & 0 & 0 \\ 0 & 10.3245 & 0 & 0 \\ 0 & 0 & 4.4474 & 0 \\ 0 & 0 & 0 & 10.5142 \end{bmatrix}. \quad (20)$$

To compare the filter performances, we set $m = 5$, $\gamma_{1,2,3,4,5} = (0.1, 0.3, 0.5, 0.7, 0.9)$ and define the probability matrix as

$$\Pi = \begin{bmatrix} 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \end{bmatrix}. \quad (21)$$

After properly tuned, all of the designed filters can now be compared in real practical situations of indoor robot localization for produced errors. To make it possible with a high confidence, we organize an experiment in two tests, which parameters are listed in Table I.

1) *Test 1*: We first assume that tuning is provided near optimally (Test 1) as described above. Accordingly, we sketch in Fig. 4 the mobile robot coordinates estimated by the filters along with the planned path and noisy UWB-based measurement data obtained using the least-square method. What follows from this experiment is that all filters produce very consistent estimates and we infer that accurate tuning makes them almost equally effective. To support this inference, we sketch the localization root mean square errors (RMSEs) in Table II and one deduces that all errors range lower the UWB-based data. It is also seen that, as expected, the cEKF filter is the most accurate as being near optimal, unlike the EFIR filters, which are unbiased.

Overall, Test 1 reveals that the adapted coloredness factor γ makes it possible to achieve the highest localization accuracy under the CMN than by the standard filters and modified filters with a constant γ .

2) *Test 2*: A situation changes dramatically when we refer to the practical inability to measure accurately the noise statistics and use parameters of Test 2 listed in Table I with permitted moderate errors. The errors estimated by the filters

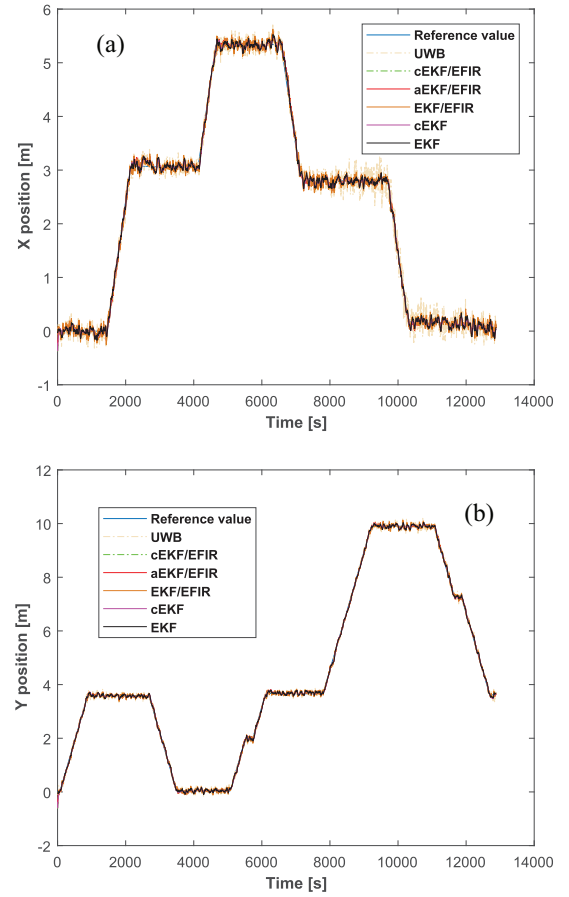


Fig. 4. Estimates of the mobile robot east coordinate x and north coordinate y provided by the filters: (a) east direction and (b) north direction.

TABLE II
LOCALIZATION RMSEs PRODUCED BY THE FILTERS IN TEST I

Filter	RMSE (m)		
	East direction	North direction	mean
UWB	0.133	0.084	0.109
EKF	0.067	0.055	0.061
cEKF	0.059	0.056	0.058
EFIR	0.082	0.064	0.073
cEKF/EFIR	0.067	0.058	0.063
aEKF/EFIR	0.070	0.058	0.064

under CMN and errors in the noise covariances are sketched in Fig. 5 and the relevant CDFs can be found in Fig. 6. What these results reveal is that even under moderate errors in the noise covariances the cEKF becomes inefficient and such that in some parts it performs worse than the UWB that is practically unacceptable. The latter means that the EKF and cEKF are highly sensitive to errors in the noise statistics. Just on the contrary, the cEKF/EFIR and aEKF/EFIR filters demonstrate the best performances to mean that efforts made in their design have resulted in end-to-end robust Algorithms 1 and 2. It is seen that the aEKF/EFIR filter is a bit worse in Test 2 than the cEKF/EFIR filter. But do not forget that γ for the cEKF/EFIR filter was found at a test stage offline, while the aEKF/EFIR filter determines γ adaptively in real time. The corresponding localization RMSEs supporting Test 2 are listed in Table III and one infers that the cEKF/EFIR and aEKF/EFIR filters are

Algorithm 2: aEKF/EFIR Algorithm

Data: $y_k, M^{cE}, N^{cE}, x_0^{cE}, P_0, Q, R$
Result: $\hat{x}_k^{cE}$

```

1 begin
2   for  $k = 1 : \infty$  do
3     for  $i = 1 : m$  do
4       if  $k < N^{cE} - 1$  then
5          $z_k = y_k - \gamma_i y_{k-1}$ ;
6          $C_k = H_k - \gamma_i H_k F^{-1}$ ;
7          $\hat{P}_k^{cE(i)} = \Delta \hat{P}_{k-1}^{cE(i)} A^T + \Theta$ ;
8          $S_k = C_k \hat{P}_k^{cE(i)} C_k^T + R + H_k \phi_k + \phi_k^T C_k^T$ ;
9         Compute  $\Lambda_k^{cE(i)}$  by (16);
10        Compute  $\rho_k^{cE(i)}$  by (17);
11         $K_k = \hat{P}_k^{cE(i)} C_k^T S_k^{-1}$ ;
12         $\hat{x}_k^{cE(i)} = A \hat{x}_{k-1}^{cE(i)} + \chi z_k$ ;
13         $\hat{x}_k^{cE(i)} =$ 
14         $\hat{x}_k^{cE(i)} + K_k [z_k - h(\hat{x}_k^{cE(i)}) + \gamma_i h(\hat{x}_{k-1}^{cE(i)})]$ ;
15         $\hat{P}_k^{cE(i)} = (I - K_k C_k) \hat{P}_{k-1}^{cE(i)}$ ;
16      else
17         $m^{cE} = k - N^{cE} + 1, s = m^{cE} + M^{cE} - 1$ ;
18         $\tilde{x}_s^{cE(i)} = \hat{x}_s^{cE(i)}$ ;
19         $\tilde{P}_s^{cE(i)} = \hat{P}_s^{cE(i)}$ ;
20         $G_s = I$ ;
21        for  $l = s + 1 : k$  do
22           $z_l = y_l - \gamma_i y_{l-1}$ ;
23           $C_l = H_l - \gamma_i H_l F^{-1}$ ;
24           $G_l = [C_l^T C_l + (F G_{l-1} F^T)^{-1}]^{-1}$ ;
25           $K_l = G_l C_l^T$ ;
26           $\tilde{x}_l^{cE(i)} = F \tilde{x}_{l-1}^{cE(i)}$ ;
27           $\tilde{P}_l^{cE(i)} = F \tilde{P}_{l-1}^{cE(i)} F^T + Q$ ;
28           $S_l = C_l \tilde{P}_l^{cE(i)} C_l^T + R + H_l \phi_l + \phi_l^T C_l^T$ ;
29          Compute  $\Lambda_l^{cE(i)}$  by (16);
30          Compute  $\rho_l^{cE(i)}$  by (17);
31           $\tilde{x}_l^{cE(i)} =$ 
32           $\tilde{x}_l^{cE(i)} + K_l (z_l - h(\tilde{x}_l^{cE(i)}) + \gamma_i h(\tilde{x}_{l-1}^{cE(i)}))$ ;
33           $\tilde{P}_l^{cE(i)} = \tilde{P}_l^{cE(i)} -$ 
34           $(2 \tilde{P}_l^{cE(i)} C_l^T + 2 \phi_l + G_l C_l^T S_l) C_l G_l$ ;
35        end for
36         $\hat{x}_k^{cE(i)} = \tilde{x}_l^{cE(i)}$ ;
37         $\hat{P}_k^{cE(i)} = \tilde{P}_l^{cE(i)}$ ;
38      end if
39    end for
40     $\hat{x}_k^{cE} = \sum_{i=1}^m \theta_k^{cE(i)} \hat{x}_k^{cE(i)}$ 
41  end for
42 end

```

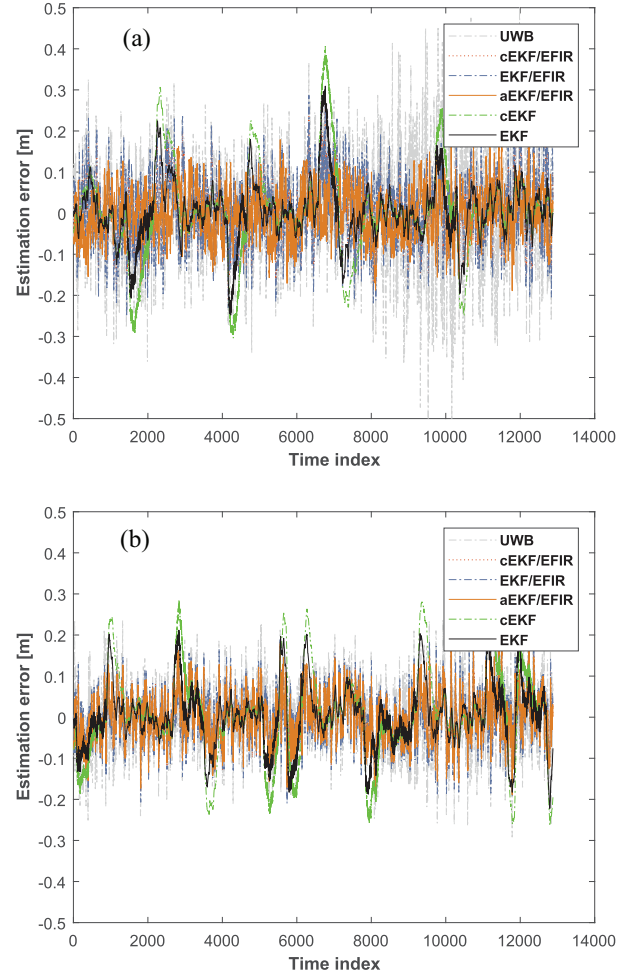


Fig. 5. Estimation errors produced by the filters under CMN and moderate errors in the noise covariances in Test 2: (a) east direction and (b) north direction.

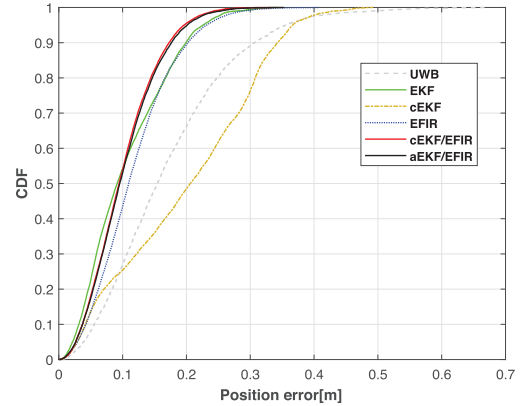


Fig. 6. Localization CDFs corresponding to errors in Fig. 5 in Test II.

the most accurate estimators for practical use in indoor robot localization.

B. Testing in Barrier Environment

In this section, we test the system taking into account the barrier flat layer, which is shown in Fig. 7. Four UWB RNs are deployed as shown in Fig. 7, and the robot moves around the concrete column. We tune the filters for $\tau = 0.1$ s and

$\sigma_w = 0.6$ m/s and specify the noise covariance Q as follows:

$$Q = \frac{(0.6 \times 0.2)^2}{9} \times \begin{bmatrix} \frac{0.02^2}{4} & \frac{0.02}{2} & 0 & 0 \\ \frac{0.02}{2} & 1 & 0 & 0 \\ 0 & 0 & \frac{0.02^2}{4} & \frac{0.02}{2} \\ 0 & 0 & \frac{0.02}{2} & 1 \end{bmatrix}. \quad (22)$$

TABLE III
LOCALIZATION RMSES PRODUCED BY THE FILTERS IN TEST II

Filter	RMSE (m)		
	East direction	North direction	mean
UWB	0.133	0.084	0.109
EKF	0.073	0.075	0.074
cEKF	0.149	0.139	0.144
EFIR	0.082	0.064	0.073
cEKF/EFIR	0.067	0.058	0.063
aEKF/EFIR	0.070	0.058	0.064



Fig. 7. Experimental set organized to provide UWB-based indoor robot localization (barrier flat layer).

We also set $m = 5$, $\gamma_{1,2,3,4,5} = (0.1, 0.3, 0.5, 0.7, 0.9)$ and define the probability matrix as

$$\Pi = \begin{bmatrix} 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \end{bmatrix}. \quad (23)$$

From the test data, we find the noise variances in the measured distances to be $\sigma_{v_{d1}^U}^2 = 5.56 \text{ m}^2$, $\sigma_{v_{d2}^U}^2 = 2.93 \text{ m}^2$, $\sigma_{v_{d3}^U}^2 = 17.07 \text{ m}^2$, and $\sigma_{v_{d4}^U}^2 = 3.20 \text{ m}^2$ and specify the measurement noise covariance $\mathbf{R}$ as follows:

$$\mathbf{R} = 9 \times \begin{bmatrix} 5.5597 & 0 & 0 & 0 \\ 0 & 2.9313 & 0 & 0 \\ 0 & 0 & 17.0726 & 0 \\ 0 & 0 & 0 & 3.1991 \end{bmatrix}. \quad (24)$$

The mobile robot coordinates estimated by the filters along with the planned path and noisy UWB-based measurement data obtained using the least-square method are shown in Fig. 8. We see that the trajectory measured by UWB has multiple overshoots, unlike the UWB solution discussed in Section IV-A. We also see that the EKF and cEKF cannot provide accurate estimates, while the hybrid EKF/EFIR, cEKF/EFIR, and aEKF/EFIR filters give estimates that range closer to the actual trajectory. This speaks in favor of the better performances of the filters designed. Fig. 9 sketches the estimation errors produced by the filters in the barrier environment, and we see that when the robot changes rapidly

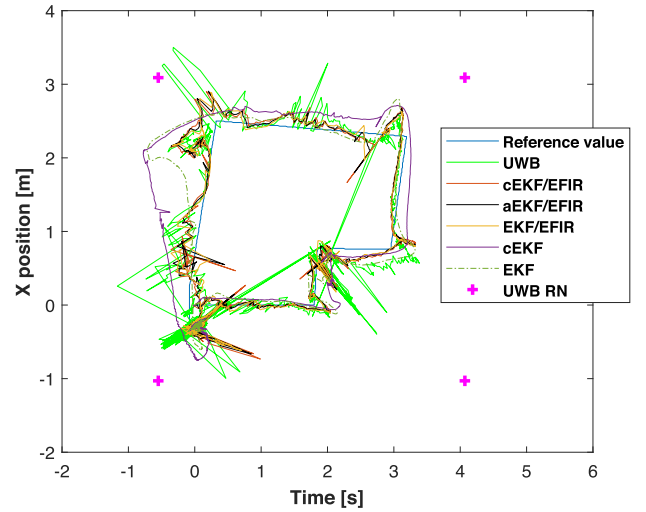


Fig. 8. Paths derived by the filters in barrier environment.

TABLE IV
PARAMETERS OF THE COMPUTER AND SOFTWARE USED IN THIS WORK

Items	Parameters
Software	Matlab R2017a
Operating system	Windows 10
CPU	Intel(R) Core(TM) i7-10875H 2.30GHz 2.30G Hz.
RAM	16GB

TABLE V
AVERAGE RUNNING TIMES OF THE FILTERS

Filtering algorithm	Running time (ms)
τ	100
EKF	0.03
cEKF	0.05
EFIR	0.60
cEKF/EFIR	0.87
aEKF/EFIR	6.18

changes the direction, the EKF and cEKF generate large prediction errors. In contrast, the EKF/EFIR, cEKF/EFIR, and the aEKF/EFIR algorithms track the trajectory more accurately. The localization CDFs corresponding to Fig. 9 are shown in Fig. 10. We see that the cEKF/EFIR has the smallest localization error when CDF is 0.9, and the proposed aEKF/EFIR filter is a bit worse than the cEKF/EFIR. Recall that optimal γ for the cEKF/EFIR filter is computed offline using all sensors data, while the proposed aEKF/EFIR filter updates γ in real time for each averaging horizon.

C. Computational Complexity

Here, we will consider the computational complexity of the proposed solution. We employ the Lenovo Savior Y9000K computer with parameters are listed in Table IV. The algorithms are implemented using MATLAB R2017a for data collected from the UWB-based localization. A special software is used to collect data for sensors, and the MATLAB R2017a is used.

The computing times of the EKF, cEKF, EFIR, cEKF/EFIR, and aEKF/EFIR over time are sketched in Fig. 11. In this figure, the computation time of the EKF is much shorter

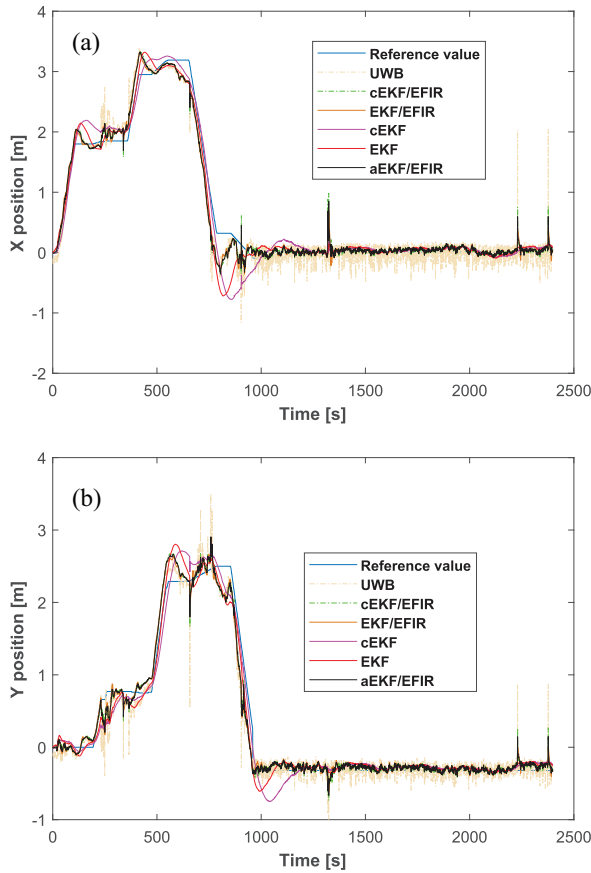


Fig. 9. Estimation errors produced by the filters in barrier environment: (a) east direction and (b) north direction.

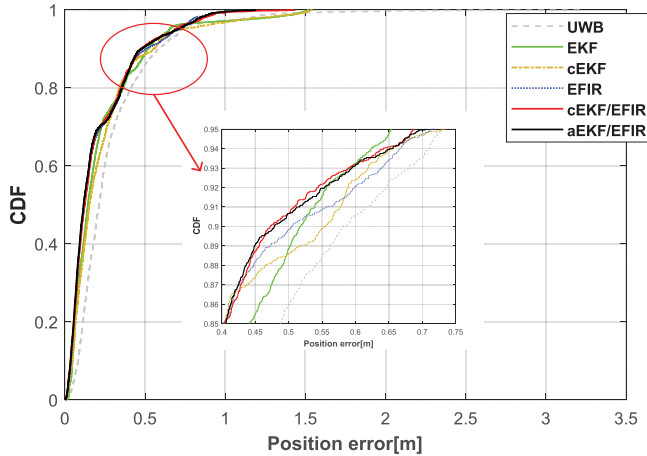


Fig. 10. Localization CDFs corresponding to errors in Fig. 9.

than the EFIR filter and the modified versions. Compared with the EFIR filter, the running time of the cEFIR filter is slightly higher, and the aEKF/EFIR filter has the highest running time. The average running times of the EKF, cEKF, EFIR, cEKF/EFIR, and aEKF/EFIR algorithms are listed in Table V, and we see that though the proposed aEKF/EFIR filter is the most time consuming, its running time is still smaller than τ used in the tests.

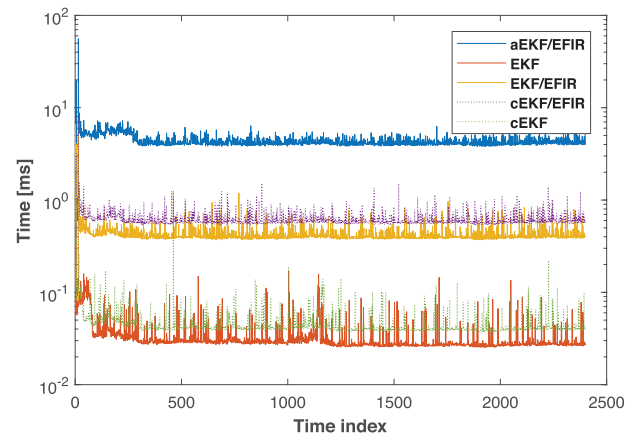


Fig. 11. Running times of the tested filters.

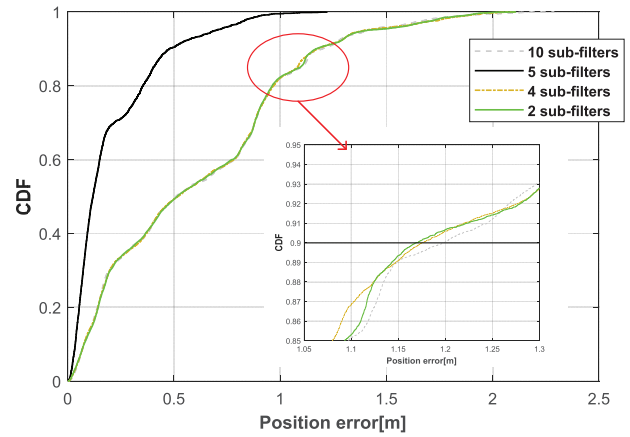


Fig. 12. Effect of the number of subfilters in the proposed filter on the localization CDF.

D. Effect of the Number of Subfilters

We finally investigate the effect of the number of subfilters on the proposed filter performance. We compare the localization CDFs obtained by the proposed filter composed with 2, 4, 5, and 10 subfilters. The results are sketched in Fig. 12, from which we see that the smallest localization error is produced when the filter is composed of five subfilters. Further increasing the number of subfilters does not give a desirable effect.

V. CONCLUSION

The cEKF/EFIR filtering algorithm modified in this article under CMN and the aEKF/EFIR algorithm adapted to nonconstant conditions through the time-varying CFM has demonstrated better performances than the basic EKF and EFIR filters and the cEKF and cEFIR filtering algorithms modified for Gauss–Markov CMN. That has become possible by designing the hybrid EKF and EFIR structures and self-determining the CFM at each time step using a filter bank operating on merged averaging horizons. Experimental testing by the mobile robot indoor localization has confirmed better performances of the hybrid filters under CMN. Since the CMN is typical for UWB-based distance measurements, the developed end-to-end cEKF/EFIR and aEKF/EFIR filtering algorithms can find applications in practical designs.

Referring to the promising results of this investigation, we are now working on further developments of the proposed algorithms for more complex industrial environments and plan to report the results in near future.

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